Eunji Kim

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PROFESSIONAL SUMMARY

Urban Data Science graduate student with a multidisciplinary background in architecture, logistics operations, and applied analytics. Six years of professional experience translating spatial data into infrastructure and business decisions across logistics, real estate, and public-sector planning. Builds end-to-end pipelines — from raw spatial and socioeconomic data through ML modeling and optimization — to surface actionable insights for cities and operators.

EDUCATION

NEW YORK UNIVERSITY, Tandon School of Engineering — *New York, NY*
Master of Science, Urban Data Science (GPA 3.9 / 4.0) — *Expected May 2027*

Relevant Coursework: Applied Data Science, Machine Learning for Cities, Big Data Management & Analysis, Geographic Information Systems

HANYANG UNIVERSITY, School of Architecture — *Seoul, South Korea*
Bachelor of Architecture — Graduated Summa Cum Laude

TECHNICAL SKILLS

- **Languages & Libraries |** Python (Pandas, GeoPandas, NumPy, scikit-learn, SciPy, PyTorch, Matplotlib, Plotly, Pydeck), SQL
- **Geospatial & Mapping |** ArcGIS Pro, ArcGIS Online, QGIS, geospatial joins, raster/vector analysis
- **Data & Cloud |** Oracle, MongoDB, Apache Spark, BigQuery exposure
- **Analytics & BI |** Tableau, Power BI, Excel (advanced), Google Workspace
- **Modeling Methods |** Random Forest, PCA, K-Means clustering, SARIMA time-series, linear/integer optimization

PROFESSIONAL EXPERIENCE

NEW YORK UNIVERSITY — *New York, NY*
Planning Intern, Strategic Assessment, Planning & Design team — *Oct 2025 – Present*

- Developed a data analysis pipeline from the university data warehouse to a Tableau internal server, streamlining spatial-data workflows for campus planning analysts.
- Built interactive dashboards aggregating disparate datasets — bridging spatial inventory with financial metrics — to derive actionable insights for capital-projects prioritization.
- Analyzed building energy-meter data with Python-based models to evaluate current utilization, identify inefficiencies, and inform retrofit and capital recommendations.

COUPANG — *Seoul, South Korea*
Designer, Space Design & Innovation, Logistics Launch team — *Jun 2023 – Jun 2025*

- Engineered spatial capacity models for 400+ last-mile delivery nodes, applying rigorous metric analysis to optimize network throughput and extend customer order cut-off times by 4 hours.
- Scaled national service coverage by 200% via geospatial feasibility analysis on 200+ candidate locations, using location intelligence to drive strategic site-selection decisions.
- Formulated space-planning standards from regional user-behavior data, rolling out delivery-experimentation protocols across 100+ operating sites to maximize operational efficiency.
- Revamped facility layouts at 8 high-risk sites to resolve capacity shortages, generating $75,000 in annualized cost savings.

GANSAM ARCHITECTS & PARTNERS — *Seoul, South Korea*
Architect — *Jan 2020 – Jun 2023*

- Led a Smart Logistics Center project end-to-end — from concept through construction documents — coordinating cross-functional teams across mechanical, electrical, and fire-safety disciplines.

- Standardized company-wide drawing guidelines and person-hour estimates, producing a system-based production workflow that improved resource-planning accuracy across project teams.

SELECTED PROJECTS

Cool Roofs: A Data-Driven Framework for Urban Heat Mitigation *May 2026*

Applied Data Science · Team of 4 · Python, scikit-learn, GeoPandas, SARIMA

- Built a PCA + K-Means clustering pipeline that classified 40,013 Manhattan buildings into bright/medium/dark roof typologies using RGB extraction from 158 orthoimagery tiles joined to PLUTO building attributes.
- Ran SARIMA time-series modeling on NASA MODIS land-surface temperature (2009–2024) to test whether the NYC Cool Roofs program produced detectable cooling at neighborhood and city scales.
- Designed a binary integer optimization model under a $24M fixed budget; two scenarios (maximize thermal impact vs. prioritize Heat Vulnerability Index) selected ~3,370 buildings, projecting a 0.78°C mean surface-temperature reduction.
- Translated findings into a policy-ready prioritization framework, replacing the current voluntary-participation program with targeted investment.

Energy Burden Prediction Model for NYC Census Tracts *May 2026*

Machine Learning for Cities · Team of 4 · Python, scikit-learn, ACS, PLUTO

- Engineered a feature set merging urban form (PLUTO), tree canopy + temperature anomaly, transit accessibility (subway/bus buffers), and ACS socioeconomic indicators across 2,200+ NYC census tracts.
- Trained Random Forest regressor on residential energy burden (% of income), tuned with RandomizedSearchCV under 5-fold CV; ran an ablation against a socio-only Linear-Regression baseline to quantify the lift from built-environment features.
- Performed K-Means clustering and feature-importance analysis to separate structural drivers (building age, FAR, residential share) from socioeconomic drivers, informing targeted retrofit and assistance policy.

Convertshare: Feasibility Assessment Tool for Office-to-Residential Conversion *Apr 2026*

ConTech Alliance Hackathon · 1st Prize · Python, financial modeling

- Developed an interactive feasibility tool targeting up to 90 million sq ft of vacant Manhattan office space, scoring conversion viability at the building level.
- Led the financial modeling workstream, building a profit-calculation framework covering acquisition, construction, and design costs to support investor and operator decision-making.

NYC Public Sports Facilities Analysis & Navigator *Dec 2025*

Big Data Management & Analysis · Python, MongoDB, Generative AI

- Built a spatial search engine for NYC public sports facilities leveraging generative AI; optimized retrieval algorithms streamlined user search experiences by 70%.

Blank Street Coffee — NYC Market Analysis (GIS) *Dec 2025*

ArcGIS Online · arcg.is/1iuifT0

- Conducted a site-selection market analysis for Blank Street Coffee in NYC using ArcGIS, layering competitor density, foot-traffic proxies, demographic and income data to surface high-potential locations.

Mapping 19th-Century New York *Dec 2025*

ArcGIS Online · arcg.is/1r0TWq1

- Reconstructed historical 19th-century New York City through georeferenced historical maps, integrating cartographic sources into an interactive ArcGIS web map.

CERTIFICATES & AWARDS

- **1st Prize** — 2026 ConTech Alliance Hackathon · *Apr 2026*
- **Registered Architect** — Korean Institute of Registered Architects (KIRA) · *Jun 2025*